Humid adiabatic atmosphere — Solution

Y22 (2022) — English translation

A1 **1.00 pts**

Find the air temperature T_{in} at a height h above the Earth's surface. Express your answer in terms of T_0, g, μ_{in}, R and the adiabatic exponent γ_{in}.

Let's consider moving a portion of air to a height h:

$$P_1 dV_1 - P_2 dV_2 = \frac{C_V dm}{\mu_{\text{in}}}(T - T_0) + dmgh$$

from where:

Answer:

$$T = T_0 - \frac{\mu_{\text{in}} gh}{C_P} = T_0 - \frac{(\gamma_{\text{in}} - 1)\mu_{\text{in}} gh}{\gamma_{\text{in}} R}$$

A2 **1.00 pts**

Find the dependence of water vapor pressure on temperature $p_v(T)$. Express your answer in terms of φ_0, p_{v0}, T_0, μ_v, μ_{in}, γ_{in} and T.

From the condition of steam equilibrium we obtain:

$$dP_{\text{v}} = -\rho_{\text{v}} g dh$$

from which:

$$\frac{dP_{\text{v}}}{P_{\text{v}}} = \frac{\mu_{\text{v}}\gamma_{\text{in}}}{\mu_{\text{in}}(\gamma_{\text{in}} - 1)} \frac{dT}{T}$$

Integrating, we find:

Answer:

$$P_{\text{v}} = \varphi_0 P_{\text{v}0} \left(\frac{T}{T_0}\right)^{\frac{\mu_{\text{v}}\gamma_{\text{in}}}{\mu_{\text{in}}(\gamma_{\text{in}}-1)}}$$

A3 **0.50 pts**

Using the graph of $p_{s.v.p.}(T)$, find for $\varphi_0 = 65\%$ and $T_0 = 287$ K the height h above the Earth's surface at which the vapor becomes saturated.

Solving the equations graphically, we find:

$$T_{\text{sat}} \approx 6{,}6°C$$

Then:

Answer:

$$h = \frac{\gamma_{\text{in}} R(T_0 - T_{\text{sat}})}{(\gamma_{\text{in}} - 1)\mu_{\text{in}} g} \approx 757 \text{ m}$$

B1 **1.00 pts**

The specific heat of vaporization depends on temperature T as follows:

$$\lambda(T) = \lambda_n + k(T - T_n)$$

Find k. Answer express through c, R, μ_n and exponent/index adiabat γ_n. Note: if you cannot solve this part, then in what follows consider/count, that $\lambda = \lambda_n = const$. This does not lead to loss of points.

The amount of heat required to evaporate mass m is equal to:

$$Q = U_{\mathrm{v}} - U_{\mathrm{liq}} + A$$

Since density water one can neglect:

$$A = pV = \frac{mRT}{\mu}$$

Since liquid is weakly compressible, the heat needed for its heating in any process is, to good accuracy, proportional to the change in its internal energy:

$$U_{\mathrm{liq}}(T_2) - U_{\mathrm{liq}}(T_1) = cm(T_2 - T_1)$$

then

$$U_{\mathrm{liq}}(T) = cmT + mC$$

where C - some constant quantity Then:

$$Q = mC + m\left(\frac{\gamma_{\mathrm{v}} R}{\mu_{\mathrm{v}}(\gamma_{\mathrm{v}} - 1)} - c\right) T = \lambda(T) m$$

Determine C from initial condition, find:

$$\lambda(T) = \lambda_n + \left(c - \frac{\gamma_{\mathrm{v}} R}{(\gamma_{\mathrm{v}} - 1)\mu_{\mathrm{v}}}\right)(T_n - T)$$

We get the answer:

Answer:

$$k = \frac{\gamma_{\mathrm{v}} R}{(\gamma_{\mathrm{v}} - 1)\mu_{\mathrm{v}}} - c$$

B2 0.50 pts

Show that when the released air mass moves, the amount of heat received by the system enclosed in its volume can be represented as follows:

$$\delta Q = m_{in}\left(\frac{\gamma_{in} R}{\mu_{in}(\gamma_{in} - 1)} dT + g dz + \lambda d\alpha\right)$$

For the amount of heat supplied to the system we have:

$$\delta Q = \frac{C_V m_{\mathrm{in}}}{\mu_{\mathrm{in}}} + P_{\mathrm{in}} dV_{\mathrm{in}} + \lambda(T) dm_{\mathrm{v}}$$

or same

$$\delta Q = \frac{C_P m_{\mathrm{in}}}{\mu_{\mathrm{in}}} dT - V_{\mathrm{in}} dP_{\mathrm{in}} + \lambda(T) dm_{\mathrm{v}}$$

From the condition equilibrium air:

$$dP_{\mathrm{in}} = -\rho_{\mathrm{in}} g dh$$

from which:

$$\delta Q = m_{\mathrm{in}}\left(\frac{\gamma_{\mathrm{in}} R}{\mu_{\mathrm{in}}(\gamma_{\mathrm{in}} - 1)} dT + g dz + \lambda(T) d\alpha\right)$$

B3 — 0.50 pts

Find $\frac{d\alpha}{dz}$. *Express your answer using* μ_v, μ_{in}, P_{in}, $P_{s.v.p.}$, $\frac{dP_{s.v.p.}}{dT}$, $\frac{dT}{dz}$ *and* $\frac{dP_{in}}{dz}$

For α, taking into account the equation of state, we have:

$$\alpha = \frac{m_\mathrm{v}}{m_\mathrm{in}} = \frac{\mu_\mathrm{v}}{\mu_\mathrm{in}}\frac{P_\mathrm{v}}{P_\mathrm{in}}$$

Differentiate:

$$\frac{d\alpha}{dz} = \frac{\mu_\mathrm{v}}{\mu_\mathrm{in}}\left(\frac{1}{P_\mathrm{in}}\frac{dP_\mathrm{v}}{dz} - \frac{P_\mathrm{v}}{P_\mathrm{in}^2}\frac{dP_\mathrm{in}}{dz}\right)$$

Consider derivative pressure pair by height as derivative complex function:

$$\frac{dP_\mathrm{v}}{dz} = \frac{dP_\mathrm{v}}{dT}\frac{dT}{dz}$$

from where:

Answer:

$$\frac{d\alpha}{dz} = \frac{\mu_\mathrm{v}}{\mu_\mathrm{in}}\left(\frac{1}{P_\mathrm{in}}\frac{dP_\mathrm{v}}{dT}\frac{dT}{dz} - \frac{P_\mathrm{v}}{P_\mathrm{in}^2}\frac{dP_\mathrm{in}}{dz}\right)$$

B4 — 1.00 pts

Find the temperature gradient $\frac{dT}{dz}$. *Express your answer in terms of* α, g, R, γ_{in}, T, μ_{in}, μ_v *and* $\lambda(T)$.

Equating δQ to zero, we find:

$$\frac{dT}{dz} = -\frac{g - \lambda\frac{\mu_\mathrm{v}}{\mu_\mathrm{in}}\frac{P_\mathrm{v}}{P_\mathrm{in}^2}\frac{dP_\mathrm{in}}{dz}}{\frac{\gamma_\mathrm{in}R}{\mu_\mathrm{in}(\gamma_\mathrm{in}-1)} + \lambda\frac{\mu_\mathrm{v}}{\mu_\mathrm{in}}\frac{1}{P_\mathrm{in}}\frac{dP_\mathrm{v}}{dT}}$$

Quantity $\frac{dP_\mathrm{v}}{dT}$ is found from the Clapeyron-Clausius equation:

$$\frac{dP_\mathrm{v}}{dT} = \frac{\mu_\mathrm{v}\lambda(T)P_\mathrm{v}}{RT^2}$$

After substitution we get:

Answer:

$$\frac{dT}{dz} = -\frac{(\gamma_\mathrm{in}-1)\mu_\mathrm{in}g}{\gamma_\mathrm{in}R}\frac{\left(1+\frac{\lambda(T)\mu_\mathrm{in}\alpha}{RT}\right)}{\left(1+\frac{\lambda(T)\mu_\mathrm{v}\alpha}{RT}\frac{(\gamma_\mathrm{in}-1)\lambda(T)\mu_\mathrm{in}}{\gamma_\mathrm{in}RT}\right)}$$

B5 — 1.80 pts

Find the temperature T_0 *on the Earth's surface.*

Let's look at the break point on the graph. At this point $\alpha = \alpha_1$. The reason for the break is the beginning of the condensation process, and therefore for the temperature gradient to the left of the break point the formula from part $A1$ is applicable, and to the right - the formula from part $B4$. Then the value of α_1 can be found from the ratio of the angular coefficients at the break point:

$$k = -\left(\frac{dT}{dz}\right)_{T\to T_1-0}\frac{\gamma_\mathrm{in}R}{(\gamma_\mathrm{in}-1)\mu_\mathrm{in}g}$$

from which:

$$\alpha_1 = \frac{1-k}{\dfrac{\lambda(T_1)\mu_\text{v}}{RT_1}\left(k\dfrac{(\gamma_\text{in}-1)\mu_\text{in}g}{\gamma_\text{in}R}\dfrac{\lambda(T_1)}{gT_1} - \dfrac{\mu_\text{in}}{\mu_\text{v}}\right)} \approx 1{,}01\cdot 10^{-2}$$

Further:

$$\alpha_1 = \frac{\mu_\text{v}p_\text{s.v.p.}(T_1)}{\mu_\text{in}p_\text{in}}$$

From the adiabat equation for dry air we have:

$$p_\text{in} = p_0\left(\frac{T_1}{T_0}\right)^{\frac{\gamma_\text{in}}{\gamma_\text{in}-1}}$$

Then we get:

Answer:

$$T_0 = T_1\left(\frac{\alpha_1\mu_\text{in}p_0}{\mu_\text{v}p_\text{s.v.p.}(T_1)}\right)^{\frac{\gamma_\text{in}-1}{\gamma_\text{in}}} \approx 309{,}2\ \text{K}$$

B6 — 0.70 pts

Estimate at what altitude h_2 snow begins to fall.

From the slope of the first section of the graph, we find the height h_1, corresponding to the break in the graph:

$$h_1 = \frac{\gamma_\text{in}R(T_0-T_1)}{(\gamma_\text{in}-1)\mu_\text{in}g} \approx 2680\ \text{m}$$

Conduct straight, correspond temperature $T_\text{pl} = 273$ K, we find:

Answer:

$$h_2 \approx 4{,}69\ \text{km}$$